Lakshmi Siri Appalaneni

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Education

University of Texas at Dallas - Bachelor of Science in Computer Science, Honors Collegium V — Aug 2025 - May 2029

GPA: 3.97/4.00 · *AES Scholar · Minor:* ***Business Intelligence & Analytics***

Coursework: Data Structures & Algorithms, Software Engineering, Systems Programming, Computer Architecture, Discrete Mathematics, Linear Algebra, Probability & Statistics in CS

Technical Skills

Languages: **C++**, **Python**, **Java**, JavaScript, TypeScript, SQL, HTML, CSS

Systems & Infrastructure: **Git**, **GitHub** (Actions, pull requests, code review), **CI/CD**, Docker, GCP, AWS, FastAPI, React.js, Node.js, Flask

Data & Machine Learning: MongoDB, PostgreSQL, DuckDB, GCS Parquet, scikit-learn, XGBoost, CatBoost, Prophet, NLTK

AI-Enabled Development: Claude Code, Cursor, GitHub Copilot, LLM code review, **prompt engineering**, agentic workflows (LangGraph, MCP)

Specializations: **Well-Architected Systems**, **Open-Source Contribution**, Root-Cause Debugging, Full-Stack Development

Experience

Reliability Design Automation Laboratory (RaDA Lab) - Undergraduate Researcher — Jan 2026 - Present

University of Texas at Dallas · Mentor: Prof. Yanwen Xu

- Improves real-world streamflow discharge prediction for hydropower reliability, cutting **Root Mean Squared Error (RMSE)** by **20-40%** via a **physics-informed digital-twin Long Short-Term Memory (LSTM)** surrogate model across **5** Western U.S. watersheds.
- Achieves **93.8-94.6%** empirical coverage against a **95%** target through statistical ablation studies isolating the physics constraint's contribution to generalization; advances a **first-author manuscript** through review at NAPS 2026.

CodePath AI 301 - Open-Source Contributor — Jun 2026 - Present

Advanced AI Engineering Program

- **Diagnosed and fixed** a production index-out-of-bounds panic in Lance's Rust search index builder, verified against **635** passing tests and merged into pull request **#7330** (awaiting review); separately reproduced a LanceDB data-corruption bug affecting **15,536** of **120,000** vector rows.

Google Developer Group (GDG) on Campus UTD - Industry Director — Jan 2026 - Present

University of Texas at Dallas, Richardson TX

- **Negotiates $12K+** in sponsorships across **6** partner organizations for HackDSC'26; translates technical scope into terms ~**350** non-technical stakeholders could act on across **3** engineering tracks.
- Cuts candidate interview prep time by roughly **40%** by engineering **Remix**, a FAISS-based **Retrieval-Augmented Generation (RAG)** pipeline (LangChain, Gemini API, SQL), shipped with a two-person team.

Nebula Labs (UTD) - Software Engineer — Sep 2025 - Present

University of Texas at Dallas, Richardson TX

- **Serves 10,000+** students by delivering full-stack features for two open-source platforms (React.js, Node.js, Go, MongoDB) across **30+**-engineer Agile sprints, **critically reviewing** AI-generated code before every GitHub pull request reaches production.

Association for Computing Machinery (ACM) Projects UTD - Backend ML Lead — Jan 2026 - May 2026

University of Texas at Dallas, Richardson TX

- **Architected SafeWay** to improve safe street navigation across Chicago, a distributed CatBoost and XGBoost hurdle model scoring crash risk at **Area Under the Curve (AUC) 0.891** and **R^2 0.745** across **2.3M+** records and **29,500** intersections.
- **Cut query latency** by roughly **80%** by redesigning the database layer to GCS Parquet with DuckDB; automated nightly **Continuous Integration/Continuous Deployment (CI/CD)** retraining via GitHub Actions on every merged pull request.

Projects

ReelSkill - Agentic Skill Extraction Pipeline | *Python, Google ADK, Gemini, FastAPI, MCP* — Jun - Jul 2026

- **Built an agent pipeline** (**Google Agent Development Kit (ADK)** 2.3, Gemini) that watches a shared Instagram Reel end to end, ingesting clips up to **19MB** natively and sampling **30-100** frames for longer videos, to extract tutorial steps and infer steps a fast-cut edit skipped.
- Packaged each result into a reusable skill file and exposed the library through a **4**-tool **Model Context Protocol (MCP)** server, built for a Kaggle x Google ADK workshop capstone.

Pondr - Knowledge Retention Platform | *Python, FastAPI, LangGraph, XGBoost, Gemini API, MongoDB* — Mar 2026

- **Models knowledge-retention curves** across **5** domains using Ebbinghaus forgetting-curve decay and XGBoost, trained on **1,000** simulated learner trajectories and integrating **7+** third-party APIs into a real-time adaptive analytics loop.

Cue - Agentic Workflow Automation | *TypeScript, React, FastAPI, Gemini, MongoDB Atlas, MCP* — Jan - Feb 2026

- **Cuts workflow setup time** from roughly **20** minutes to under **1** minute via a Chrome extension that predicts a user's next **5** tasks and autonomously triggers actions across **9+** Google Workspace tools via MCP.

HarmoniQ - Network Health Platform | *Python, Flask, scikit-learn, Prophet, React, TypeScript* — Nov 2025

- **Predicted network outages** at **87-92%** accuracy by extracting sentiment from **100K+** social posts and live telemetry, replacing reactive incident response with automated alerting.

Recognition & Honors

- **Y Combinator AI Startup School 2026** - Selected from **30,000** global applications (invite-only) — Jul 2026
- **1st Place** - HackAI'26 (AI Track), Pondr — Mar 2026
- **1st Place** - ACM Research Symposium, post-quantum information security — Nov 2025
- **3rd Place** - HackUTD'25 (T-Mobile Challenge) + MLH Best Use of Auth0, HarmoniQ — Nov 2025
- **ICPC AlgoQueen'23** - Gold Medalist, Global Rank **5** (C++) — May 2023